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course

ML, DL, NLP

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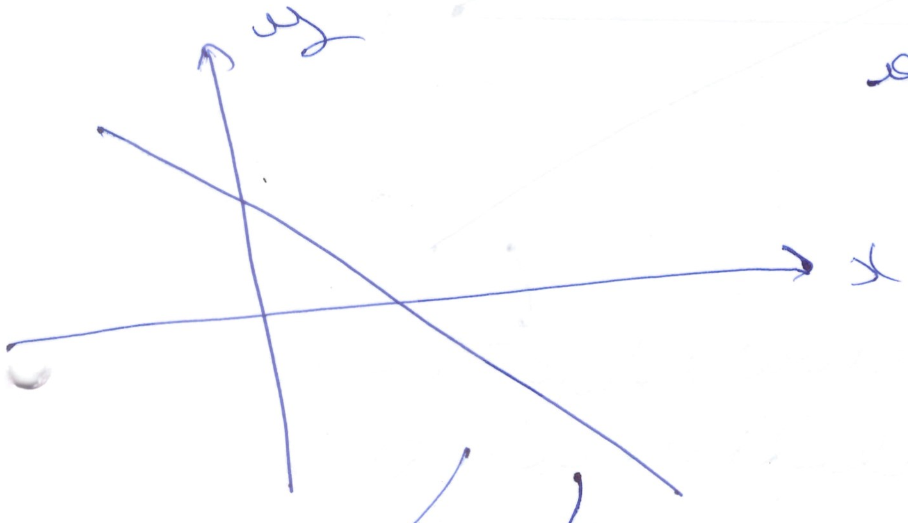
Sec-30 SVM

Lec-3 SVM maths Intuition

Lec-4 SVC cost Function

- Support vector machines (SVC) maths Intuition



$ax + by + c = 0$

$w^T x + b = 0$

Here, we need to create a best fit line

$ax + by + c = 0$

$\Downarrow$

$w_1 x_1 + w_2 x_2 + b = 0$

$w_1x_1 + w_2x_2 + b = 0$

$w_1x_1 + w_2x_2 + b = 0$

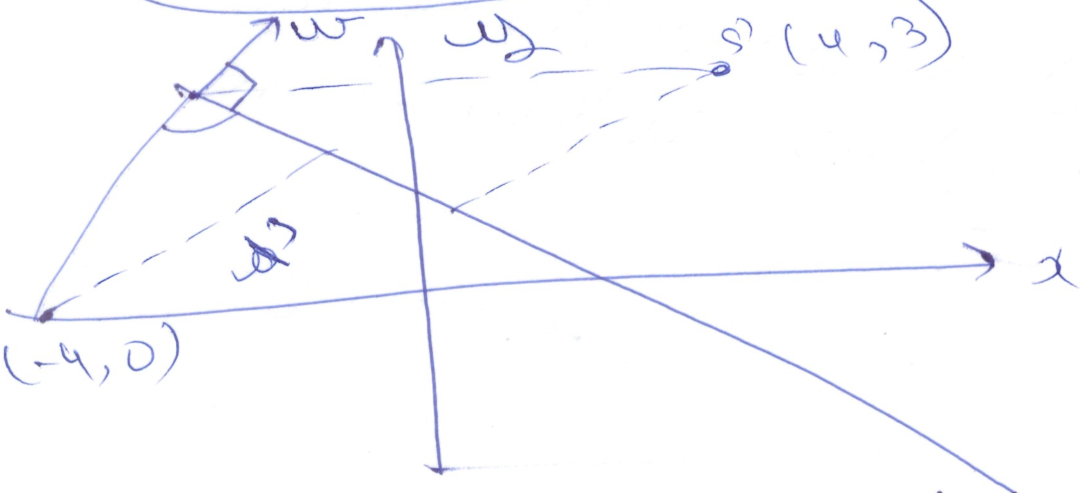
$w^Tx + b = 0$

$\Downarrow$

If line passes through origin

$b = 0$

$w^Tx = 0$

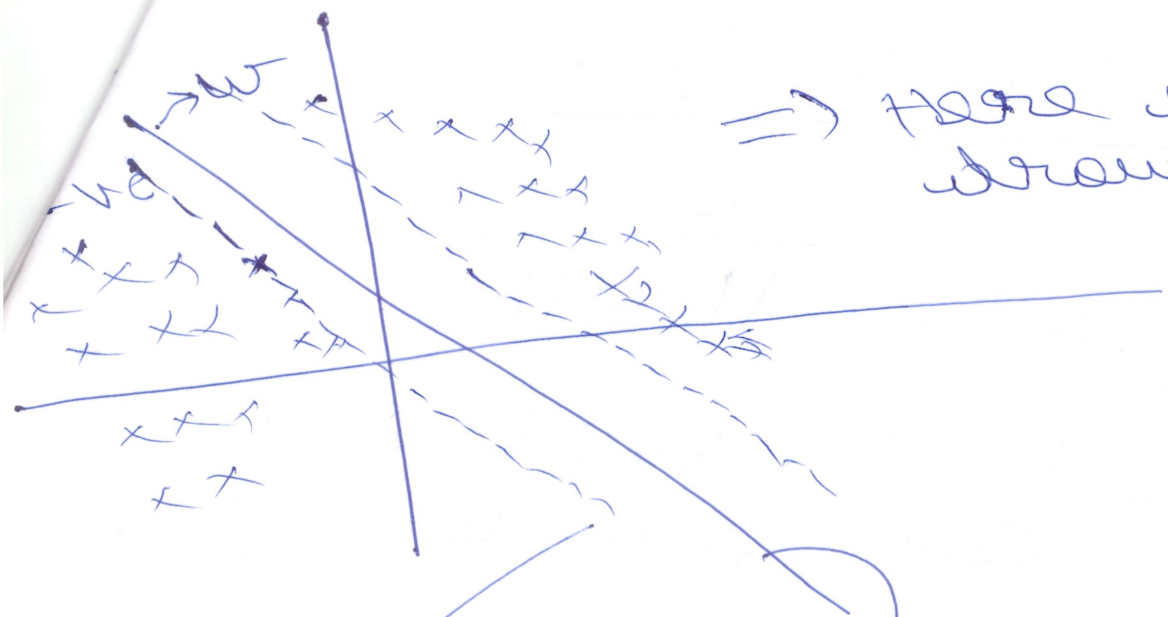


It will always ~~be~~ $w^Tx + b = 0$ be -ve below plane

d = -ve below plane

d = +ve above plane.

Basically we need to have the maximum distance b/w marginal planes



⇒ Here we have drawn the

$$w^T x + b = 0$$

$$w^T x + b = -1$$

$$w^T x_1 + b = 1$$

$$w^T x_2 + b = -1$$

$$w^T (x_1 - x_2) = +2$$

Unit vector -- Magnitude of the vector 'w'

$$w^T x + b = -1$$

⇒ **Cost Function :-**

$$\underset{w,b}{\text{maximize}} \ \frac{2}{\|w\|}$$

→ This is basically the distance b/w marginal planes

constraint such that

$$y_i \begin{cases} +1 & w^T x + b \geq 1 \\ -1 & w^T x + b \leq -1 \end{cases}$$

correctly classified points

For all correct points

constraint → $y_i * (w^T x + b) \geq 1$

The required constraint

$$y_i * (w^T x + b) \geq 1$$

For all correct points

constraint → $y_i * (w^T x + b) \geq 1$

maximize$_{(w, b)}$ $\frac{2}{\|w\|}$ $\Rightarrow$ $\boxed{\min_{(w, b)} \frac{\|w\|}{2}}$

cost function ~~of~~ SVM (SVC)

$$\min_{w, b} \frac{\|w\|}{2} + C_i \sum_{i=1}^{n} \xi_i$$

$\Downarrow$

Hinge Loss

How many points we want the avoid misclassification ~~ato~~ items?

$$\min_{w, b} \frac{||w||}{2} + c_i \sum_{i=1}^{n} \xi_i$$

ξ_i basically says
is the summation
from $i = 1$ to n
summation all distance of
incorrect data points from
marginal plane.